
Clustering-Based Lightweight Architecture for Image Captioning

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Abstract

Recent advances in vision–language models have significantly improved image captioning performance; however, existing systems often produce generic or semantically shallow descriptions, particularly when trained on large and non-uniform datasets. In this work, we propose a clustering-based framework for image captioning that organizes image–caption pairs into semantically coherent groups prior to model training. We then train a cluster-conditioned caption generation model that leverages cluster identity as additional contextual information, enabling more specialized and context-aware caption generation. We evaluate our approach on the Microsoft COCO dataset with BLEU, ROUGE-L, and METEOR across 3 models: ViT-GPT2, GIT, and BLIP. The results show that clustering does not consistently improve overall performance compared to training on the full dataset, with mixed results depending on the model and metric. However, when we look more closely at individual examples, we find that clustering can help generate more specific and relevant captions in certain cases. The code is available at <https://github.com/quyenltran/image-captioning>.

1 Introduction

Recent advances in vision–language modeling have significantly improved image captioning. Early methods used encoder–decoder frameworks, where convolutional neural networks (CNNs) extract visual features and recurrent neural networks (RNNs), such as LSTMs, generate captions [24, 25]. Attention mechanisms and Transformer-based architectures further improved performance by modeling relationships between visual and textual modalities [25]. More recent approaches, such as ClipCap [14] and BLIP [11], leverage pretrained representations to achieve strong results.

Despite this progress, captioning models often generate generic or vague descriptions, particularly for complex scenes [1]. A key challenge is that models are trained on large and diverse datasets, where semantically different images are treated uniformly, limiting their ability to produce specialized and context-aware captions [1, 10, 5]. Large vision–language models, including Flamingo [2] and CoCa [26], as well as newer systems like GPT-5 [16], Gemini 3 Pro [6], Gemma 3 [23], and Qwen2.5-VL [3], demonstrate strong performance but are computationally expensive and prone to inconsistent outputs [1]. This motivates exploring more efficient methods to improve caption quality without relying on large-scale models.

In this work, we propose a clustering-based framework that groups semantically similar image–caption pairs before training. Prior work suggests that multimodal clustering can improve representation learning [17, 22, 27]. By organizing data into coherent clusters, our approach aims to enable more specialized caption generation. We evaluate this method by comparing clustered and global training using standard metrics and qualitative analysis. The main contributions of this work can be summarized as follows:

1. We provide an empirical analysis of clustering-based preprocessing for image captioning and evaluate its impact on performance.
2. We present a comparative study of clustering-based and global training across multiple models, highlighting the strengths and limitations of clustering.

2 Methodology

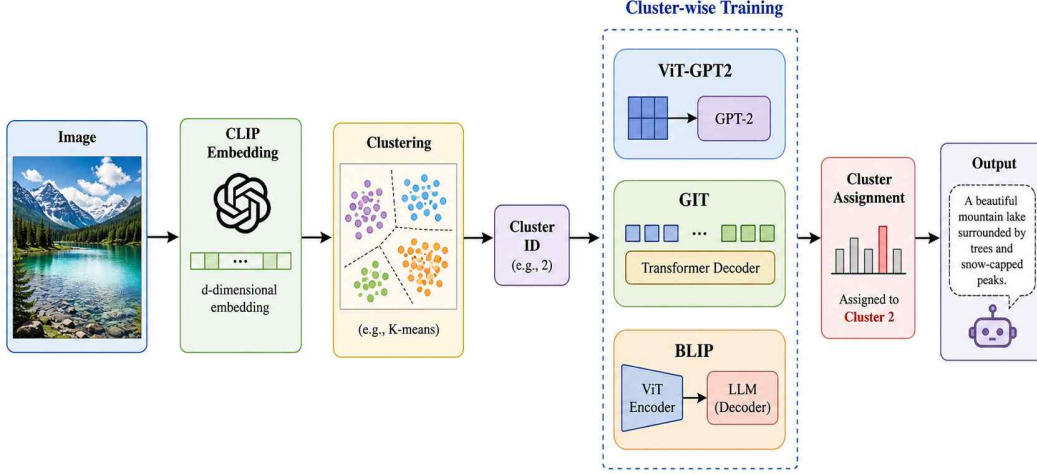


Figure 1: Overview of the proposed clustering-based image captioning framework.

We propose a clustering-based framework for image captioning that partitions the training data into semantically related groups and trains cluster-specific captioning models (Figure 1). This design is motivated by the observation that images from different semantic categories often require different captioning behaviors. For example, aviation images, indoor scenes, and transportation scenes contain distinct visual structures and descriptive patterns. Rather than training a single global captioning model, the framework specializes in caption generation according to the semantic structure of the embedding space.

Step 1 (Visual Representation). The pipeline begins by extracting semantic image embeddings using a pretrained vision encoder. Given an image I , we compute a feature representation $z_I = f_v(I) \in \mathbb{R}^d$, where f_v denotes the visual encoder. In our implementation, we use the CLIP visual encoder to obtain semantically meaningful image representations.

Following common preprocessing practices in representation learning and clustering-based methods, we adopt statistical standardization using feature-wise z -score normalization to standardize the embedding space prior to clustering [9, 19]. The normalization is computed using the empirical mean and standard deviation estimated from the training set. The encoder is kept frozen throughout all experiments to preserve the pretrained semantic structure of the embedding space and ensure consistency across clustering methods.

Step 2 (Clustering). The standardized embeddings are partitioned into semantically related groups using unsupervised clustering methods. We evaluate multiple clustering algorithms, including K-means [8], Spherical K-means [7], Hierarchical Clustering [15], Gaussian Mixture Models (GMM) [21], and representation-aware clustering approaches. Each method partitions the embedding space into K clusters $\{\mathcal{C}_1, \dots, \mathcal{C}_K\}$, where each image embedding receives a cluster assignment $k_i \in \{1, \dots, K\}$. Figure 2 illustrates representative visualizations of the clustering methods evaluated in this work.

Clustering Methods – Visual Comparison

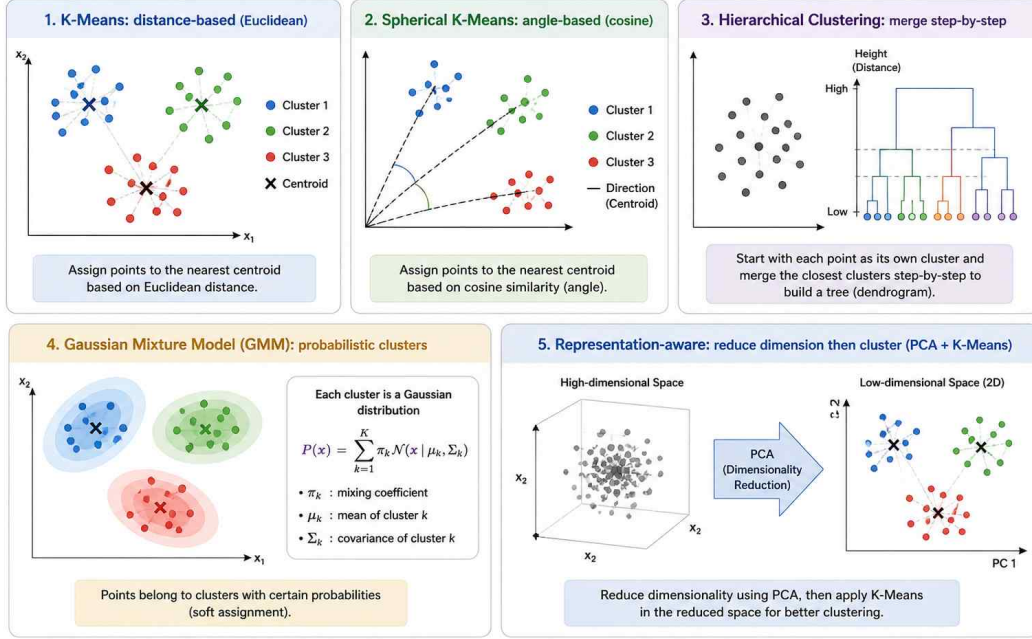


Figure 2: Visualization of the clustering methods evaluated in the proposed framework. Different clustering algorithms partition the visual embedding space into semantically related image groups.

To determine the number of clusters, we evaluate candidate values of K using repeated clustering runs and silhouette-score analysis. For each candidate value, we compute the empirical mean and standard deviation of the Silhouette Score across repeated runs and define the stability-adjusted score $\text{Score}(K) = \bar{S}(K) - \sigma_S(K)$, where $\bar{S}(K)$ and $\sigma_S(K)$ denote the empirical mean and standard deviation of the Silhouette Score, respectively. The value of K with the highest stability-adjusted score is selected for subsequent experiments.

Step 3 (Cluster-Conditioned Training). After clustering, the training dataset is partitioned into cluster-specific subsets $\mathcal{D}_k = \{(I_i, t_i) : k_i = k\}$, where t_i denotes the target caption associated with image I_i . For each cluster, we independently train a separate image captioning model using only the training samples assigned to that cluster. All cluster-specific models use the same architecture and training configuration. The only difference between models is the subset of training samples associated with each cluster. This design allows us to explore whether semantically coherent image groups lead to different captioning behaviors and whether cluster-specific training can influence caption generation quality.

Step 4 (Inference). During inference, an input image is first mapped into the embedding space using the same feature extraction and normalization procedures used during training. The image is then assigned to a cluster according to the selected clustering method. For centroid-based methods, routing is performed using nearest-centroid assignment $k = \arg \min_j \|\tilde{z}_I - \mu_j\|_2$. The corresponding cluster-specific model is then used to generate the final caption $t = g_k(I)$.

3 Experiments

3.1 Experimental Setup

Datasets. We use Microsoft COCO [13], a widely used dataset for large-scale object detection, segmentation, and captioning¹. It contains over 330K images (more than 200K labeled), approximately

¹<https://huggingface.co/datasets/lmms-lab/COCO-Caption>

1.5M object instances (e.g., person, car, bicycle, dog, airplane) across 80 object categories and 91 scene-context categories (e.g., sky, road, grass, wall, water), and 5 human-annotated captions per image. Its high-quality human annotations and multi-caption supervision align with our objective. Most captions in this dataset are approximately 8-12 words, with a relatively small number of longer descriptions exceeding 15 words. This relatively narrow range suggests that captions in the dataset tend to provide concise descriptions of visual scenes.

Data Preprocessing. All images are resized to 224×224 pixels and normalized using the pre-processing pipeline associated with CLIP ViT-B/32 [20]. Image features are extracted using the pretrained CLIP visual encoder, producing 512-dimensional representations. Captions are processed using the CLIP tokenizer based on byte-pair encoding (BPE). For each image, we retain a single caption and remove samples with missing text. No additional text normalization or filtering is applied.

Training Details. We evaluate three image captioning architectures: ViT-GPT2, GIT-base, and BLIP-base. For each architecture, we train: (i) a global model using the full training dataset, and (ii) cluster-specific models trained independently on the corresponding cluster subsets. All models are trained using identical optimization settings and decoding procedures to ensure fair comparison across experiments.

Training is performed using the HuggingFace Transformers framework with teacher-forced autoregressive caption generation and standard cross-entropy language-modeling loss. Optimization is performed using the AdamW optimizer with learning rate 5×10^{-5} and weight decay 0.01. The learning rate is kept constant throughout training. During inference, captions are generated using beam search decoding with 4 beams. Beam search maintains multiple candidate token sequences during generation and selects the sequence with the highest overall likelihood score, aiming to produce more coherent and contextually consistent captions than greedy decoding.

Due to computational limitations in training multiple cluster-specific models across multiple clustering configurations and architectures, all experiments are trained for 1 epoch. The goal of this work is therefore not to fully optimize captioning performance, but rather to investigate the relative impact of clustering-based specialization under a controlled experimental setting.

Evaluation Setup. We evaluate model performance using standard automatic captioning metrics, including BLEU [18], ROUGE-L [12], and METEOR [4], which measure n-gram overlap, sequence-level similarity, and semantic alignment between generated captions and reference captions. In addition to quantitative evaluation, we manually inspect a subset of generated captions to analyze qualitative differences between globally trained and cluster-specific models. The qualitative analysis focuses on caption specificity, contextual relevance, and common failure behaviors across different clustering configurations.

3.2 Clustering Configuration

Clustering Methods. Clustering is performed on standardized CLIP visual embeddings, as described in Section 2. We evaluate multiple clustering algorithms, including K-means, Spherical K-means, Hierarchical Clustering, and Gaussian Mixture Models (GMM).

Selection of Number of Clusters. For each clustering method, the number of clusters is searched over the range $K \in \{2, 3, \dots, 15\}$. Cluster selection is performed using the stability-based statistical clustering-selection procedure described in Section 2.

4 Results and Analysis

4.1 Analysis of Clustering Behavior

Selection of Number of Clusters. For each clustering method, the number of clusters is selected independently using the stability-adjusted silhouette criterion described in Section 3.2. Table 1 summarizes the selected value of K for each clustering method together with representative semantic categories observed within the resulting clusters. Additional clustering visualizations, silhouette analyses, PCA projections, and representative cluster examples are provided in Appendix A.

Clustering Method	Selected K	Representative Categories
K-means	4	Indoor scenes, motorcycles, airplanes, and street scenes.
Spherical K-means	5	Motorcycles, indoor scenes, airplanes, object-centered images, and street scenes.
Hierarchical Clustering	2	Mixed visual scenes together with a small outlier-like cluster.
GMM	10	Animals, bathrooms, kitchens, motorcycles, airplanes, and outdoor scenes.
Representation-aware Clustering	2	Mixed transportation scenes, indoor scenes, and object-centered images.

Table 1: Selected number of clusters and representative semantic categories across clustering methods.

The selected values of K vary substantially across clustering methods, indicating that different clustering algorithms partition the same CLIP embedding space in different ways. K-means and Spherical K-means select moderate values of K , separating the embedding space into several broad semantic categories such as indoor scenes, transportation-related images, and aviation scenes. In contrast, GMM selects a larger value of K and produces more detailed semantic groupings. Hierarchical and representation-aware clustering both select $K = 2$, partitioning the embedding space into two larger groups compared to the other clustering methods.

Overall, the selected clustering configurations suggest that the CLIP embedding space contains meaningful semantic structure that can be partitioned in multiple ways depending on the assumptions of the clustering algorithm. However, the selected values of K should be interpreted relative to the candidate search range considered in this work. Since the experiments only evaluate K over a finite range, the selected configurations represent the best-performing values within the tested range rather than globally optimal choices. Expanding the search range or using alternative clustering-selection criteria may produce different clustering configurations and semantic partitions.

4.2 Main Quantitative Results

Overall Performance Comparison. We compare globally trained captioning models against cluster-specific training across three image captioning architectures: ViT-GPT2, GIT-base, and BLIP-base using the evaluation metrics described in Section 3.1. Table 2 summarizes the performance of global and clustered training across all clustering methods. Additional metric comparison graphs are provided in Appendix B.

Model	Setting	Method	BLEU	ROUGE-L	METEOR
ViT-GPT2	Global	none	0.155	0.415	0.380
	Clustered	kmeans	0.150 (-0.005)	0.413 (-0.002)	0.384 (+0.004)
	Clustered	spherical-kmeans	0.134 (-0.024)	0.396 (-0.021)	0.369 (-0.013)
	Clustered	hierarchical	0.125 (-0.033)	0.396 (-0.021)	0.387 (+0.006)
	Clustered	GMM	0.109 (-0.049)	0.386 (-0.031)	0.359 (-0.022)
	Clustered	rep-aware	0.135 (-0.023)	0.398 (-0.019)	0.376 (-0.006)
GIT-base	Global	none	0.005	0.303	0.191
	Clustered	kmeans	0.008 (+0.003)	0.304 (+0.001)	0.189 (-0.002)
	Clustered	spherical-kmeans	0.013 (+0.007)	0.309 (-0.013)	0.204 (-0.006)
	Clustered	hierarchical	0.006 (-0.000)	0.315 (-0.006)	0.200 (-0.010)
	Clustered	GMM	0.021 (+0.015)	0.327 (+0.006)	0.214 (+0.004)
	Clustered	rep-aware	0.007 (+0.001)	0.308 (-0.013)	0.198 (-0.012)
BLIP-base	Global	none	0.157	0.437	0.414
	Clustered	kmeans	0.153 (-0.004)	0.439 (+0.002)	0.377 (-0.037)
	Clustered	spherical-kmeans	0.151 (-0.006)	0.431 (-0.008)	0.371 (-0.045)
	Clustered	hierarchical	0.160 (+0.003)	0.441 (+0.002)	0.418 (+0.002)
	Clustered	GMM	0.154 (-0.003)	0.441 (+0.002)	0.372 (-0.044)
	Clustered	rep-aware	0.148 (-0.009)	0.429 (-0.010)	0.363 (-0.053)

Table 2: Full comparison of global and clustered training across all clustering methods. Values in parentheses denote the change relative to global training (clustered – global).

Overall, clustering-based specialization does not consistently improve aggregate captioning performance. The observed effects vary across both architectures and clustering methods, with different clustering configurations producing improvements for some models and metrics while reducing performance for others.

For ViT-GPT2, the globally trained model achieves BLEU, ROUGE-L, and METEOR scores of 0.155, 0.415, and 0.380, respectively. Among the evaluated clustering methods, K-means produces the most comparable performance to global training, slightly improving METEOR from 0.380 to 0.384 while producing small decreases in BLEU and ROUGE-L. In contrast, GMM clustering results in larger reductions across all three metrics.

BLIP-base exhibits a similar pattern. The globally trained model achieves scores of 0.157 (BLEU), 0.437 (ROUGE-L), and 0.414 (METEOR). Hierarchical clustering slightly improves all three metrics, increasing BLEU to 0.160, ROUGE-L to 0.441, and METEOR to 0.418. However, several other clustering methods reduce performance, particularly for METEOR.

Among the evaluated architectures, the GIT-based shows the largest variation across clustering configurations. The globally trained model achieves BLEU, ROUGE-L, and METEOR scores of 0.005, 0.303, and 0.191, respectively. GMM clustering produces the largest improvements, increasing BLEU to 0.021, ROUGE-L to 0.327, and METEOR to 0.214. Other clustering methods produce smaller and less consistent changes across metrics.

Comparison Across Clustering Methods. Across the evaluated clustering methods, K-means and Hierarchical Clustering generally produce the most stable performance relative to global training. In particular, K-means maintains performance close to the global baseline for both ViT-GPT2 and BLIP-base, while Hierarchical Clustering achieves small improvements for BLIP-base across all evaluation metrics.

GMM produces the largest variation across architectures. While it substantially improves the performance of GIT-base, it also leads to larger metric reductions for ViT-GPT2 and BLIP-base. This behavior may reflect the larger number of clusters selected by GMM, which partitions the embedding space into more detailed semantic groups and therefore changes the distribution of training samples assigned to each cluster-specific model.

Spherical K-means and representation-aware clustering generally produce results close to, but slightly below, the corresponding global baselines across most architectures and metrics. Although these methods preserve some semantic structure within the embedding space, the resulting specialization does not consistently translate into improved aggregate captioning performance.

4.3 Qualitative Analysis of Generated Captions

Comparison Between Global and Clustered Models. To better understand the effect of clustering beyond aggregate evaluation metrics, we manually inspect captions generated by globally trained and cluster-specific models across multiple validation examples. In this comparison, the cluster-specific models correspond to the K-means clustering configuration. We focus on K-means because it is one of the most widely used and commonly studied clustering methods, making it a natural baseline for qualitative comparison across architectures. In addition, the selected K-means configuration produces multiple semantically interpretable clusters while maintaining a moderate number of clusters, allowing clearer analysis of how clustering influences caption-generation behavior. Figure 3 presents representative examples from ViT-GPT2, GIT-base, and BLIP-base, including cases where clustering improves the generated caption, produces minimal qualitative change, or introduces incorrect details. Additional qualitative examples are provided in Appendix C.

Across all three architectures, cluster-specific training frequently produces captions that are more visually specific or contextually detailed than those generated by globally trained models. This behavior is particularly visible for images belonging to semantically coherent categories such as airplanes, motorcycles, bathrooms, kitchens, and street scenes. In such cases, the cluster-specific models often generate more precise object descriptions, scene attributes, or environmental context.

For example, several clustered models explicitly identify scene structure or object relationships omitted by the globally trained baselines. In the ViT-GPT2 examples, the clustered model identifies a “car stopped at a red light at night” rather than simply describing a “traffic light sitting on the side of a

road.” Similarly, clustered BLIP examples often introduce more contextual scene information, such as public bathrooms, restaurant environments, or indoor scene composition.

At the same time, the qualitative improvements remain inconsistent across examples and architectures. In many cases, globally trained and cluster-specific models generate highly similar captions with only small wording differences. This observation is consistent with the quantitative results, where clustering frequently produces relatively small changes in aggregate evaluation metrics.

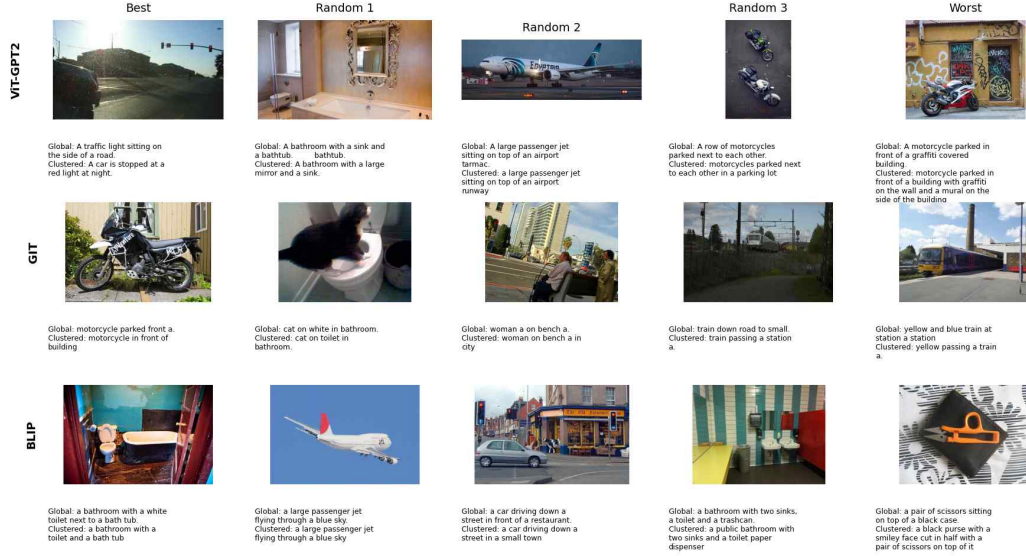


Figure 3: Qualitative comparison between globally trained models and K-means cluster-specific models across ViT-GPT2, GIT-base, and BLIP-base. The examples include successful cases, randomly selected examples, and failure cases.

Comparison Across Clustering Methods. We further compare captions generated using different clustering methods within the same captioning architecture. Figure 4 presents representative examples using ViT-GPT2 across K-means, Spherical K-means, Hierarchical Clustering, GMM, and representation-aware clustering.

We focus on ViT-GPT2 because it produces relatively interpretable sentence structure while also exhibiting noticeable qualitative variation across clustering methods. Compared to GIT-base, which often generates shorter and less stable captions, and BLIP-base, which tends to produce more consistent outputs across clustering methods, ViT-GPT2 provides clearer qualitative differences that make the effect of clustering easier to analyze.

The examples reveal that different clustering methods influence caption-generation behavior in different ways. K-means and Spherical K-means frequently produce similar captions, often generating more complete object descriptions or scene-level context compared to the global baseline. Hierarchical Clustering occasionally produces broader object-level descriptions with less contextual detail, while GMM sometimes introduces additional scene attributes or object relationships not present in the global prediction.

At the same time, several clustering methods also introduce incorrect or incomplete details. In some examples, captions become overly biased toward dominant semantic patterns within a cluster, leading to repetitive phrasing, missing objects, or hallucinated scene attributes. For example, in the kitchen example shown in Figure 4, several clustering methods generate incomplete phrases such as “kitchen a stove oven a and,” while the globally trained model produces a more grammatically coherent description. Similarly, some clustering configurations omit important contextual details or generate overly generic captions despite correctly identifying the main objects in the image. These observations suggest that cluster-specific training may reinforce common captioning patterns within each cluster, especially when the cluster contains visually similar or repetitive examples.

Overall, the qualitative analysis suggests that clustering changes caption-generation behavior even when the aggregate evaluation metrics remain relatively similar. The effectiveness of clustering-based specialization, therefore, depends not only on the captioning architecture but also on how different clustering methods partition the semantic structure of the embedding space.



Global: blue and toilet on display a
kmeans: white blue and white sitting a on white with white.
spherical_kmeans: white blue and white sitting a on white with white.
hierarchical: vases on display a display a.
gmm: blue and white on display a.
representation_aware: blue and white vase on display a.



Global: A kitchen filled with lots of wooden cabinets.
kmeans: A kitchen with a cabinet and a refrigerator.
spherical_kmeans: A kitchen with a table and a refrigerator.
hierarchical: A kitchen filled with wooden cabinets and a wooden table counter, a sink.
gmm: a kitchen filled with wooden cabinets and wooden tables and chairs.
representation_aware: A kitchen with a wooden table and wooden cabinets.



Global: kitchen a and stove oven
kmeans: kitchen a stove oven a and.
spherical_kmeans: kitchen a stove oven a and.
hierarchical: kitchen appliances a and stove oven
gmm: kitchen with white and wood.....
representation_aware: kitchen a stove oven a and.

Figure 4: Qualitative comparison across clustering methods using the ViT-GPT2 captioning architecture.

5 Conclusion

This work explored a clustering-based framework for image captioning that groups semantically similar image-caption pairs before training. Using CLIP embeddings and multiple clustering strategies, we investigated whether semantic clustering could improve caption generation across ViT-GPT2, GIT, and BLIP. Experimental results on the Microsoft COCO dataset show that clustering does not consistently improve aggregate performance across BLEU, ROUGE-L, and METEOR. While some clustering methods produced small improvements for specific models or metrics, the gains remained limited and inconsistent. In several cases, cluster-specific training reduced performance compared to global training, particularly when clusters contained mixed semantic content or limited training diversity.

The qualitative analysis nevertheless showed that clustering can influence caption-generation behavior in meaningful ways. When images aligned well with the semantic structure of their assigned clusters, cluster-specific models often generated more detailed or context-aware captions than globally trained baselines. At the same time, failure cases demonstrated that simple clustering methods such as K-means struggle to capture the continuous and overlapping nature of visual semantics, occasionally reinforcing misleading visual-text patterns and producing inaccurate captions.

This work also has several limitations. The experiments were conducted under limited training budgets, with all models trained for a single epoch across multiple clustering configurations. In addition, the study relies primarily on automatic evaluation metrics, which may not fully capture qualitative differences in caption quality or semantic grounding. Future work may explore more advanced clustering strategies, soft assignment methods, stronger conditioning mechanisms, joint representation learning approaches, and human evaluation protocols to better understand how clustering influences caption generation behavior.

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A Additional Clustering Analysis

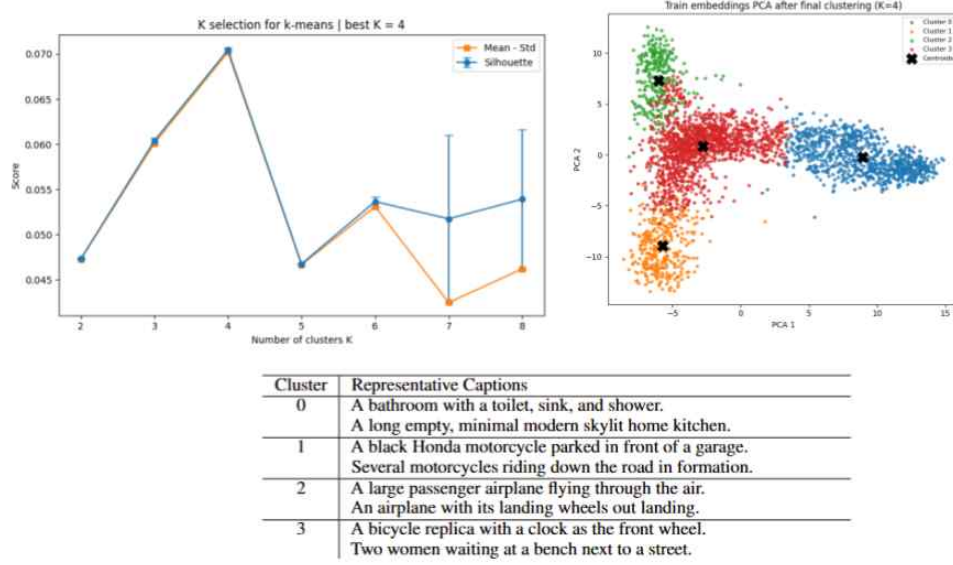


Figure 5: Detailed K-means clustering analysis.

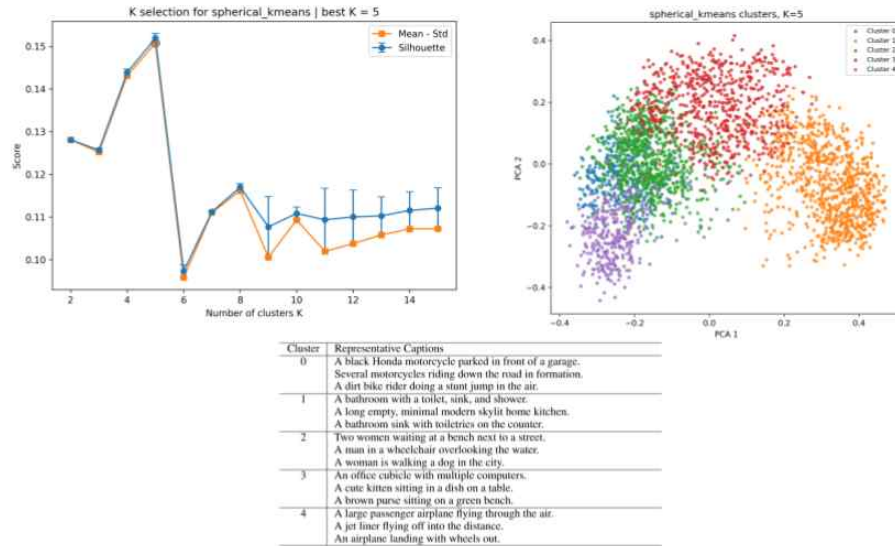
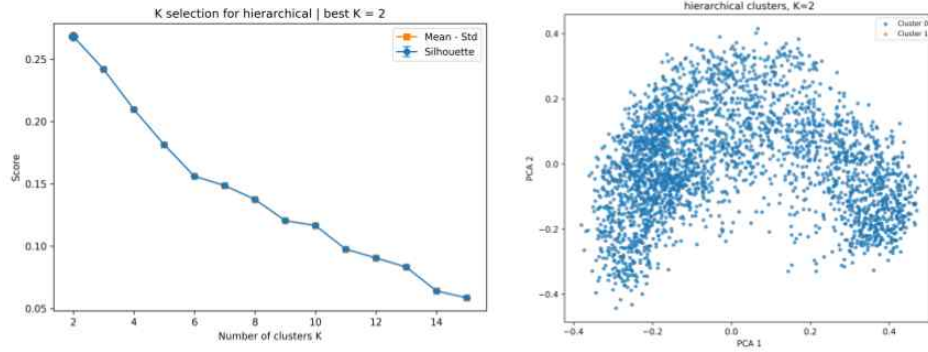


Figure 6: Detailed Spherical K-means clustering analysis.



Cluster	Representative Captions
0	A bicycle replica with a clock as the front wheel. A black Honda motorcycle parked in front of a garage. A bathroom with a toilet, sink, and shower.
1	A close up of a TV with a man wearing a suit and shirt. (single outlier cluster)

Figure 7: Detailed Hierarchical Clustering analysis.

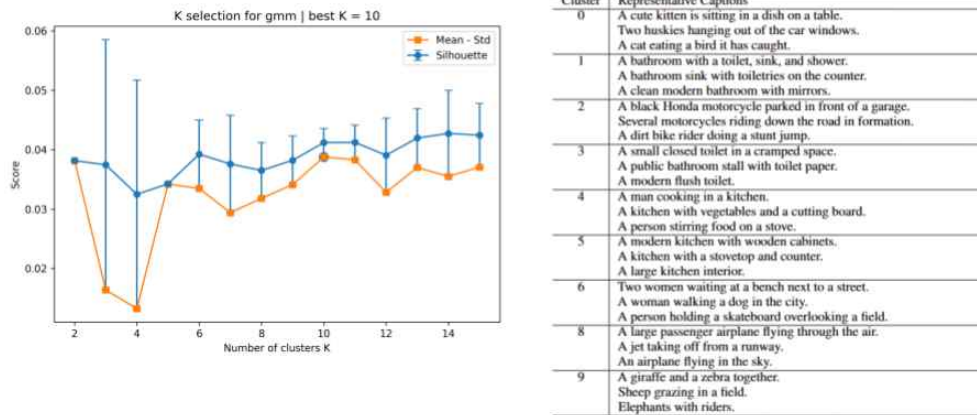


Figure 8: Detailed Gaussian Mixture Model clustering analysis.

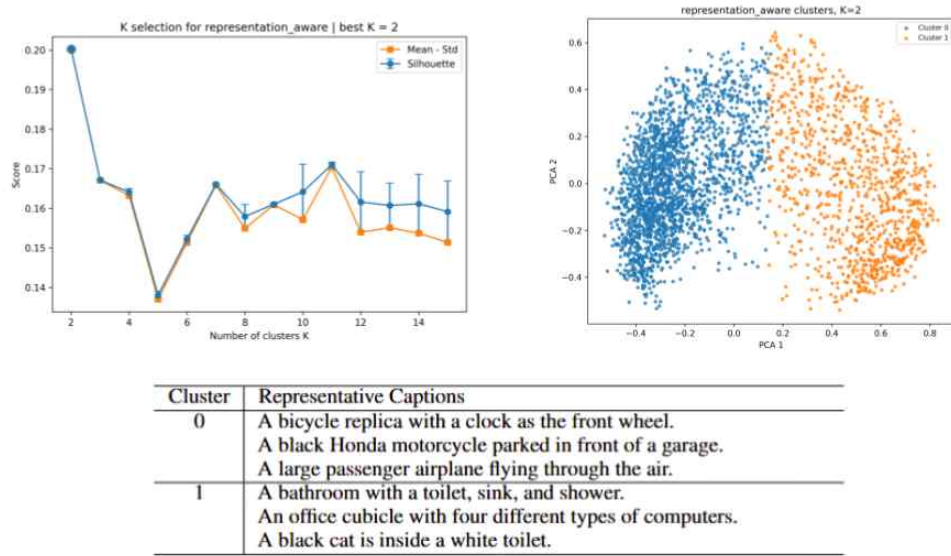


Figure 9: Detailed representation-aware clustering analysis.

B Additional Quantitative Results

B.1 Architecture-specific Metric Trends

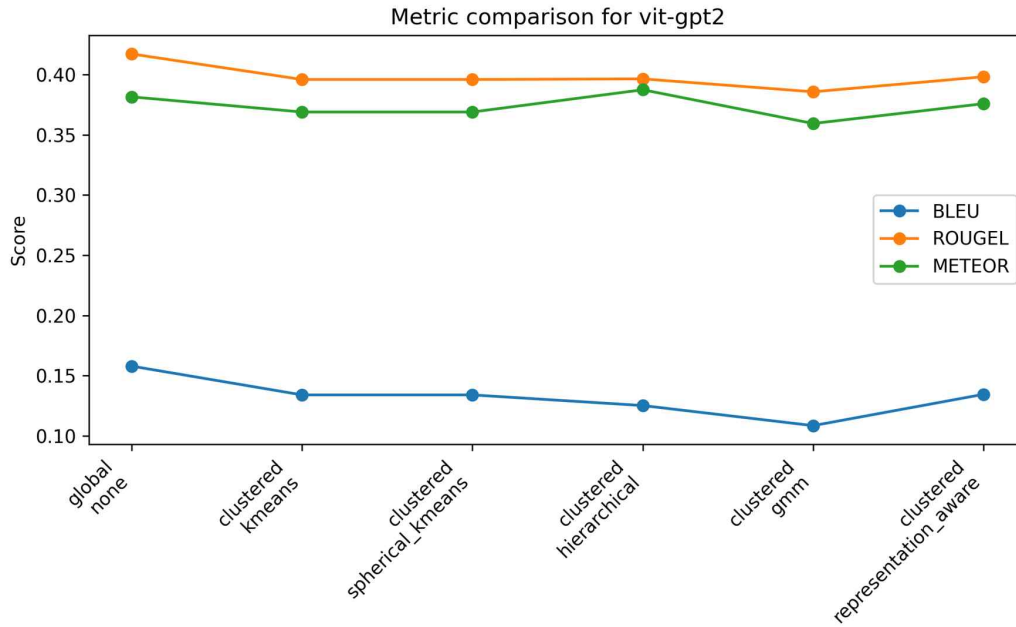


Figure 10: Metric comparison across clustering methods for ViT-GPT2.

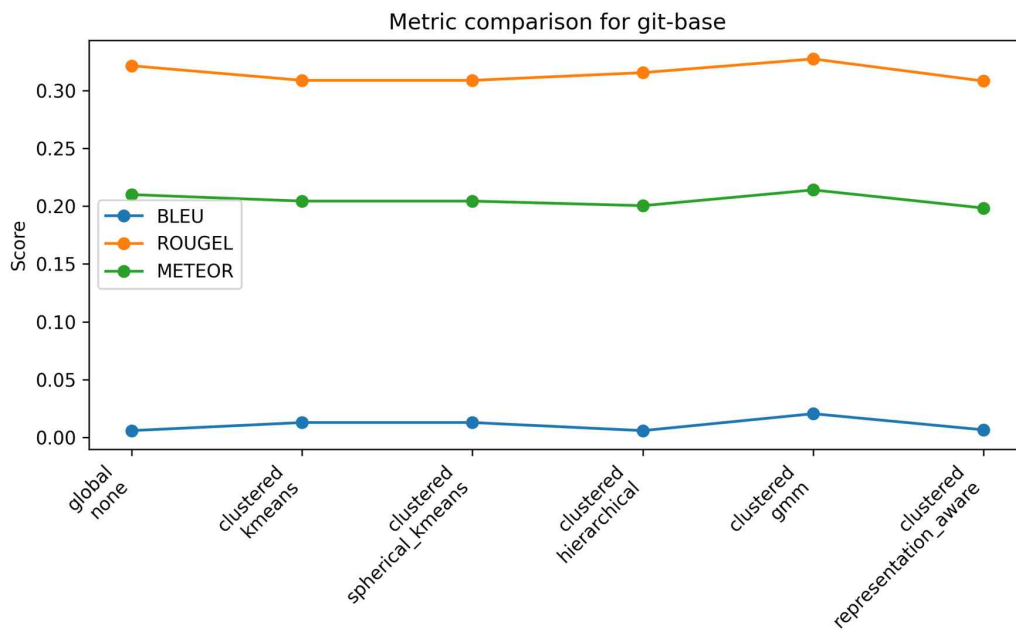


Figure 11: Metric comparison across clustering methods for GIT-base.

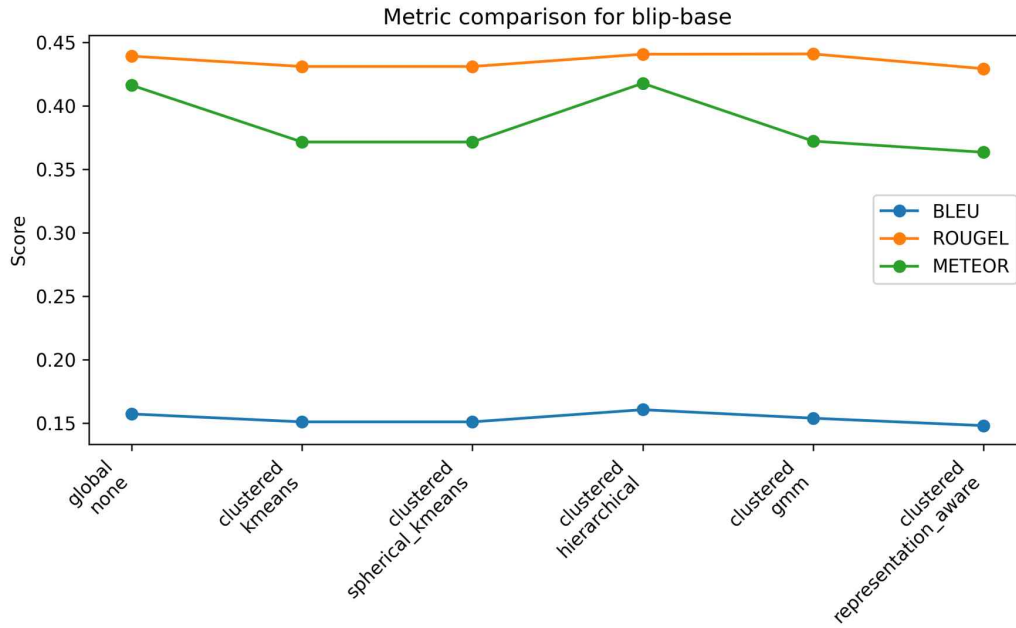


Figure 12: Metric comparison across clustering methods for BLIP-base.

B.2 Metric Difference Analysis

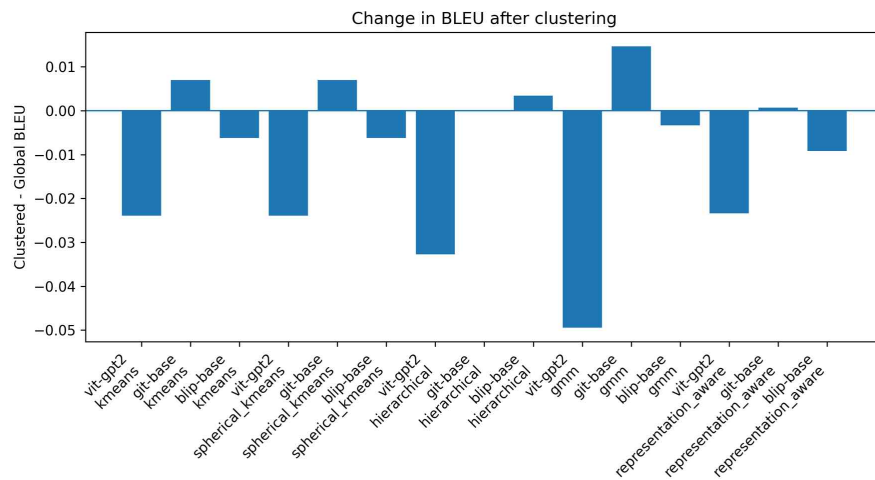


Figure 13: Difference in BLEU scores between clustered and globally trained models.

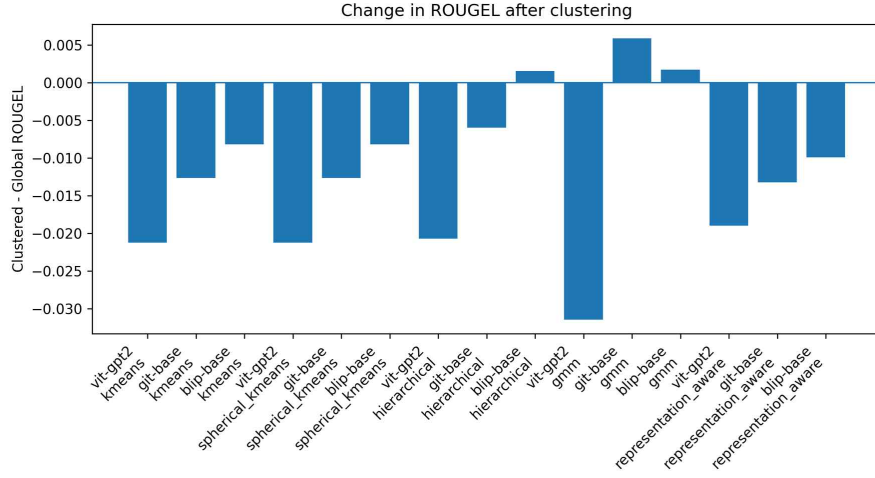


Figure 14: Difference in ROUGE-L scores between clustered and globally trained models.

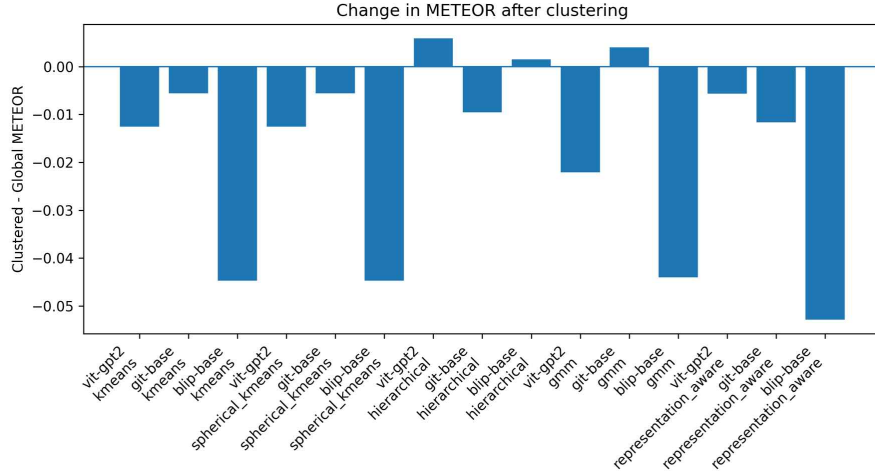


Figure 15: Difference in METEOR scores between clustered and globally trained models.

C Additional Qualitative Examples

C.1 Model-wise Qualitative Comparisons



C.2 Clustering-method Comparisons

Global vs clustered captions: kmeans

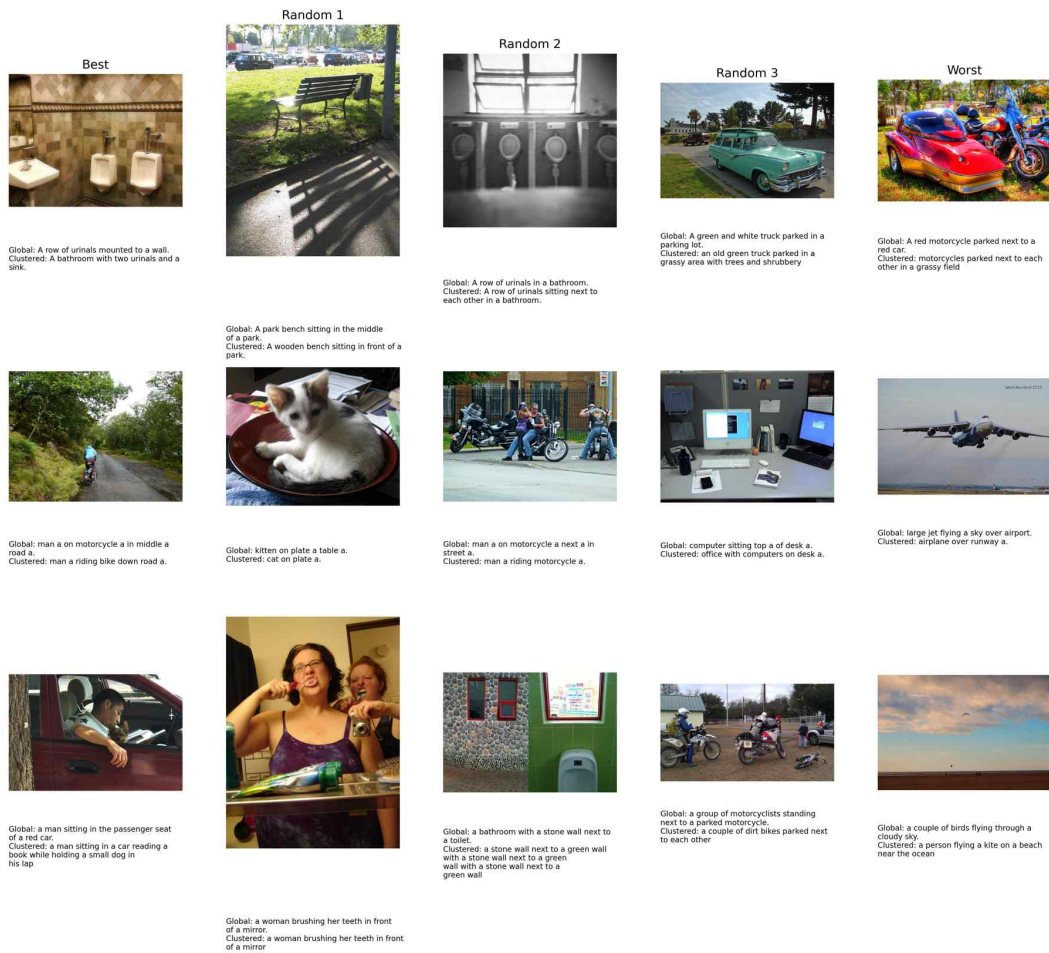


Figure 19: Additional qualitative examples for K-means clustering.

Global vs clustered captions: spherical_kmeans

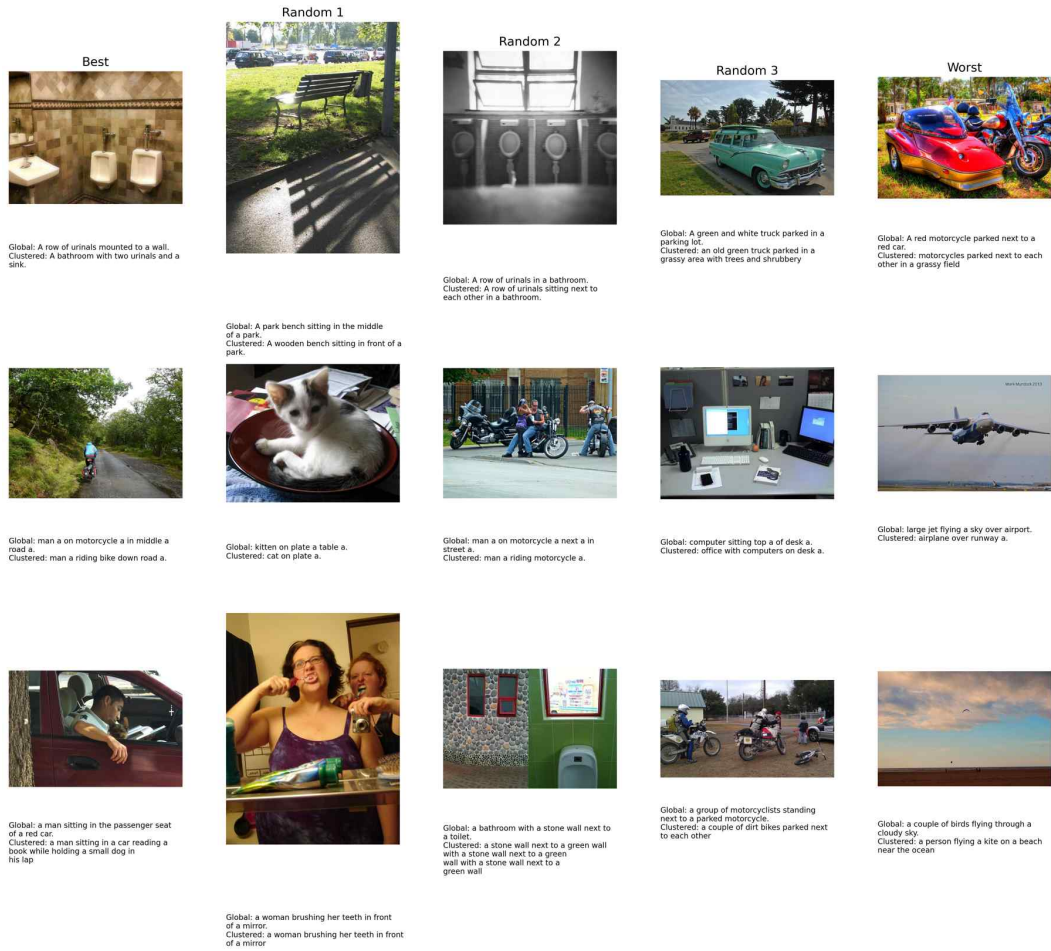


Figure 20: Additional qualitative examples for Spherical K-means clustering.

Global vs clustered captions: hierarchical

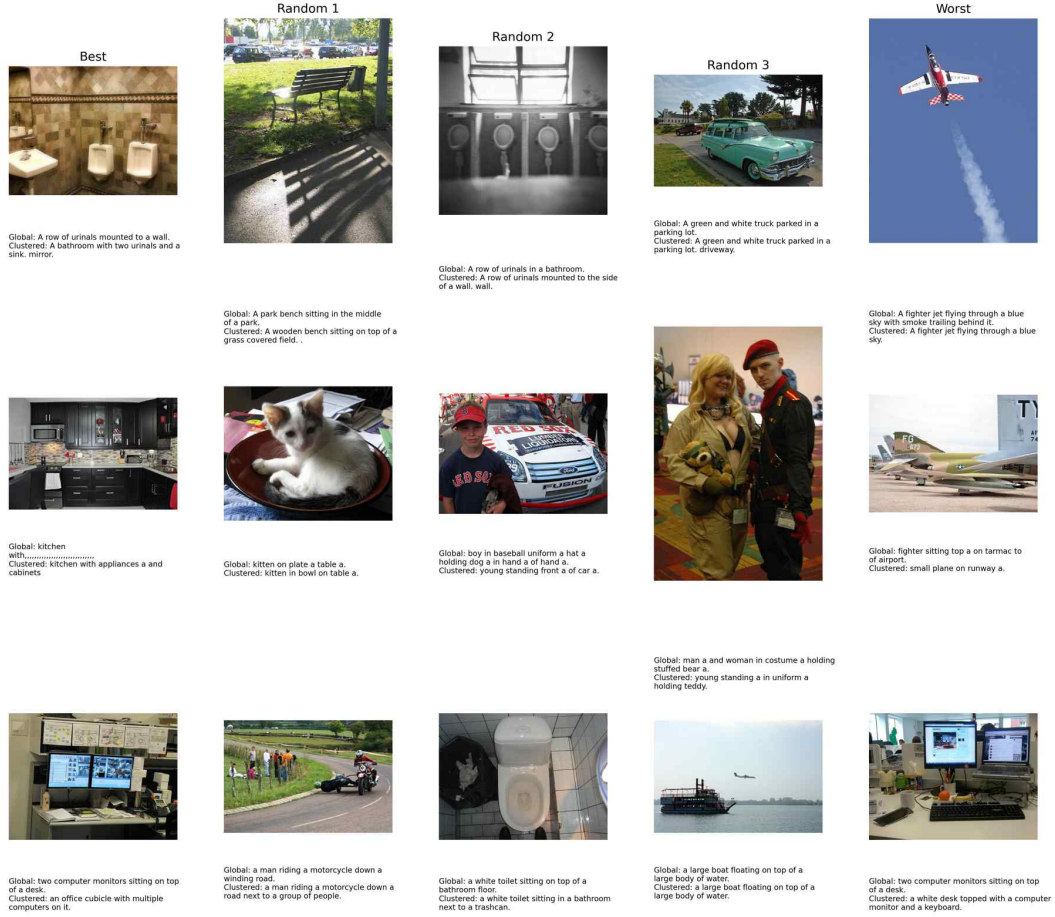


Figure 21: Additional qualitative examples for Hierarchical Clustering.

Global vs clustered captions: gmm

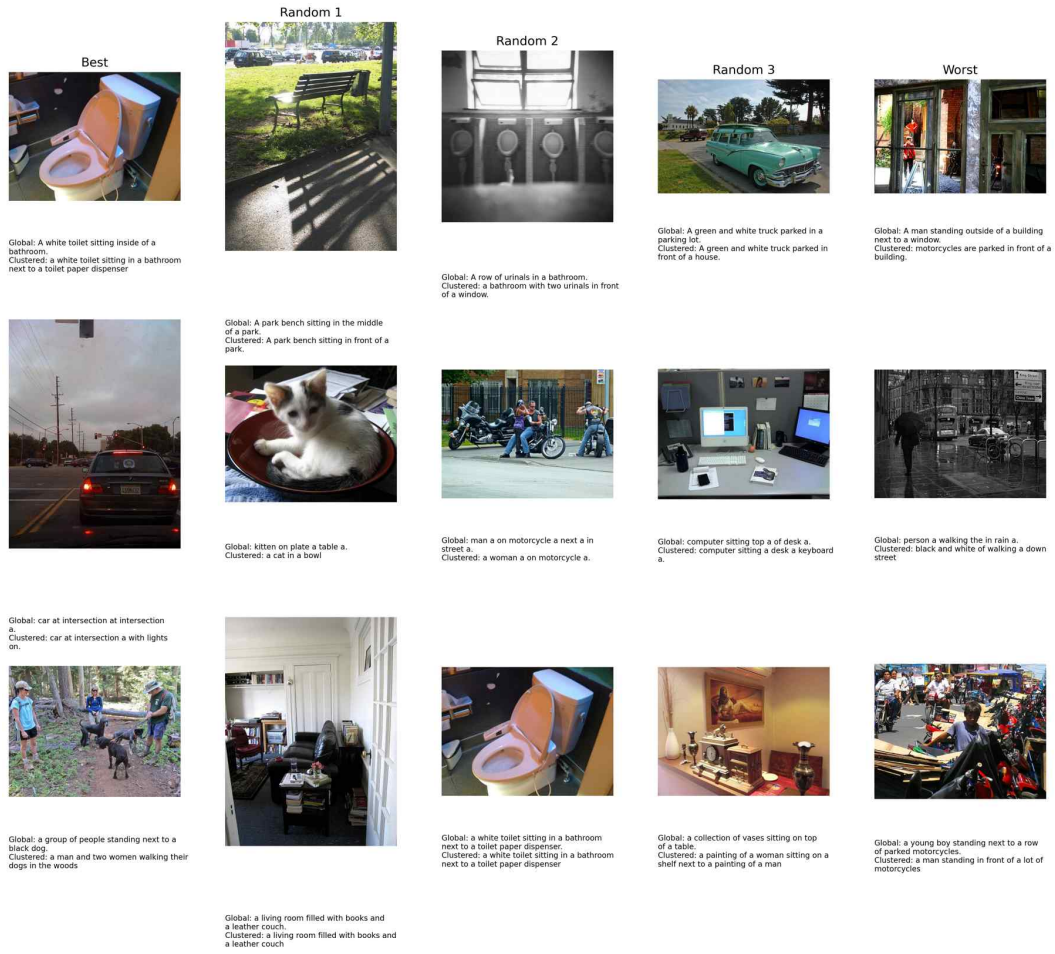


Figure 22: Additional qualitative examples for GMM clustering.

Global vs clustered captions: representation_aware

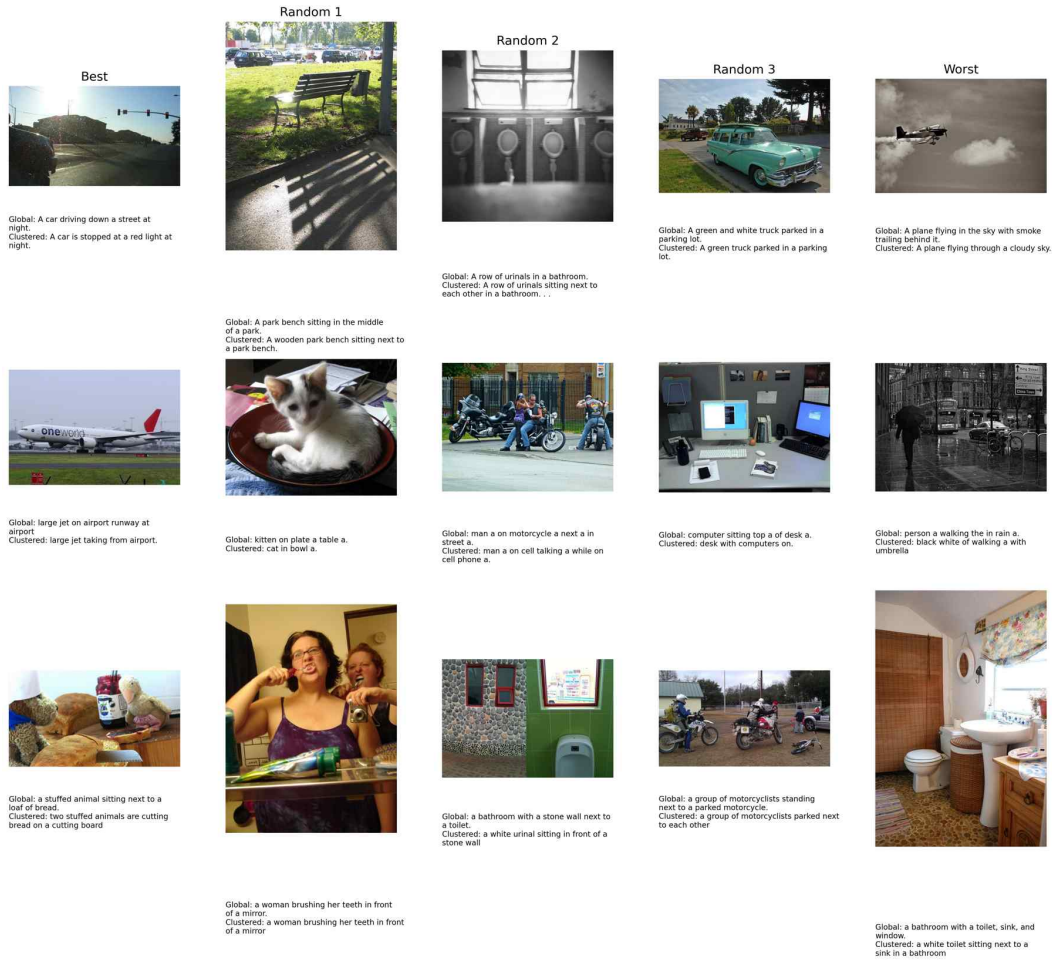


Figure 23: Additional qualitative examples for representation-aware clustering.